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Hot corrosion behavior of porous nickel-based alloys containing molybdenum in the presence of NaCl at 750 °C

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ABSTRACT

The hot corrosion of porous Ni-23Cr-xMo (0%, 4.5%, 9.0%, 13.5%, mass fraction) alloys tested at 750 °C under cyclic procedure was investigated in order to elucidate the effect of Mo addition on hot corrosion in the presence of NaCl. The hot corrosion experiments were performed at 750 °C in air with 4 mg cm^{-2} NaCl deposit. The performance of the alloys was evaluated by the results of weight change kinetics. X-ray diffraction (XRD), scanning electron microscopy (SEM), and energy dispersive X-ray spectroscopy (EDS) were used to characterize the corrosion products. The results indicate that NaCl accelerated the oxidation of the alloys by chloridized elements Mo and Cr. Among the porous Ni-23Cr-xMo alloys, Ni-23Cr-9Mo alloy exhibited the best hot corrosion resistance due to the formation of NiO-$NiCr_2O_4$-Cr_2O_3 oxide scales. Furthermore, these oxide scales were confirmed more effective to protect the alloys after adding of Mo.

1. Introduction

Naval aircrafts are always exposed to the marine environment where salty fog and high humidity are concentrated during their parking time, which could account for > 90% the service life [1,2]. Generally, NaCl enters the turbines as an aerosol and deposits on the blades and coatings. Solid NaCl at higher temperatures accelerate the hot corrosion of the turbines [3–6].

Corrosion damage is the second most frequent observed damage in aircraft structures except for fatigue cracks, especially for some parts which are easily corroded, such as the porous abradable coatings [7,8]. The abradable coatings are mostly composed of metal phase, self-lubricating nonmetal phase and a defined level of porosity. However, the abradable coatings are easily destroyed due to the pores provide penetrating paths for corrosion medium and formed galvanic cells [9,10].

Ni-based alloys are mainly used in aircraft engines as structural or coating materials due to their excellent high temperature strength and corrosion resistance [11,12]. Ni-based alloy coating is a kind of middle-temperature sealing coating, which has been widely applied in the aircraft turbine compressor casing [1]. Both Cr and Mo are important alloying elements in Ni-based alloys to improve mechanical properties by solid solution hardening. They seem to exert a synergistic effect together, especially on localized corrosion. Many researchers reported that Ni-based alloy coating containing Cr exhibits good against high temperature corrosion because of the selective oxidation of Cr to form stable and protective Cr_2O_3 [13–15]. The elements Mo is also an important element to resistant corrosion environment [16]. However, Mo enters the fused salt and forms a corrosive liquid phase of Na_2SO_4-MoO_3 in the

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Table 1
Chemical composition of various alloys (wt%).

Alloy	Ni	Cr	Mo
A1	Bal.	23	0
A2			4.5
A3			9
A4			13.5

presence of Na_2SO_4, which can lead to alloy-induced acidic fluxing of Ni [17]. Moreover, Mo is also found to have a negative effect on the oxidation at elevated temperature [18]. However, the fundamental understanding to explain this behavior is still lacking, which is a significant issue to be solve in the present research.

Compared with dense mass materials, coatings with a porous multi-phase structure are easy to be corroded. In this paper, the corrosion on a simulate coating structure of porous Ni-23Cr-xMo (0%, 4.5%, 9.0%, 13.5%, mass fraction) alloys caused by NaCl at 750 °C was studied to evaluate the effect of Mo. The reason why we test at 750 °C was based on the service temperatures (~760 °C [19]) of the alloys and NaCl melting point.

2. Experimental

2.1. Preparation of the porous materials

The Ni-based porous alloys for the present study were prepared by mixing ultrafine Ni (> 99.9% pure, 5–8 μm), Cr (> 99.9% pure, 1–2 μm), Mo (> 99.9% pure, 1–2 μm) and NH_4HCO_3 powders. The NH_4HCO_3 was added as the pore former. The chemical compositions of these alloys are reported in Table 1. The contents of Mo in A1, A2, A3 and A4 alloys were 0%, 4.5%, 9.0% and 13.5% respectively, which were used the experience of haynes625 and hastelloyC-22 for reference [16]. The evenly mixed powders were pressed into plates with a size of 35.0 × 25.0 × 3.0 mm^3 under pressure of 200 MPa for 1 min. Then the plates were sintered at 1000 °C for 8 h under flowing H_2. The porosity was 10 ± 2% which was estimated by measuring the cross-sectional images of them with software Image Pro Plus 6.0. The samples for corrosion experiments were cut into 20.0 × 10.0 × 2.0 mm^3 blocks.

2.2. NaCl-induced hot corrosion tests

Cyclic hot corrosion tests were performed in static air after NaCl was deposited on the surfaces of the samples at 750 °C. The purpose of cyclic tests was to create severe conditions for hot corrosion under repeatedly changed temperature, which simulates the service environment of the alloys [20]. At least three samples were tested to confirm the data reproducibility.

First, all the samples were mechanically grinded on SiC papers up to 1000 grit, and were ultrasonically cleaned in ethanol and acetone. And then the samples were placed in alumina crucibles and dried at 110 °C for 1 h. After that, they were weighed and sprayed with aqueous saturated NaCl solution to ensure at least 4 $mg{\cdot}cm^{-2}$ NaCl on the surfaces. Second, the samples were treated in a muffle furnace kept at 750 °C for 10 h and cooled to room temperature. Then they were washed in boiling water for dissolving the residual salts. Finally, the cleaned samples were dried and weighed by a physical balance with a sensitivity of 10^{-4} g to determine the mass change after each cycle. Before the samples were brought back into the furnace, the same amount of fresh NaCl (i.e. 4 $mg{\cdot}cm^{-2}$) was sprayed on the surface of the samples in order to a new cycle of the corrosion test was begun. The cycle was repeated for 15 times (i.e. 150 h).

The microstructure and morphology of the corrosion products were characterized by scanning electronic microscopy (SEM, JSM-7001F) and energy dispersive X-ray spectroscopy (EDS, INCA X-MAX). Phase identifications of the corrosion products were carried out by X-ray diffractometer (XRD, PANalytical B.V., X'Pert Pro MPD) using Cu Kα radiation. Thermodynamics calculations were performed with the HSC Chemistry 6.0 software in order to reveal corrosion mechanisms. The kinetics of corrosion was determined from the weight change measurements. And the weight change rate of the samples (Δw) was calculated according to the following formula:

$$\text{Weight change rate } \Delta w = \frac{\text{weight change}}{\text{sample surface area}} \tag{1}$$

3. Results and discussions

3.1. Weight change kinetics

The corrosion kinetics of the alloys have been studied by mass gain measurements. The weight change kinetics of the alloys after 15 cycles are shown in Fig. 1. In the case of NaCl corrosion, the weight losses of A1, A2 and A3 alloys as a function of corrosion time were quite obvious. In contrast, the A4 alloy did not lose its weight even after 150 h. One can see that slight mass gains of the A3 and A4 alloys were observed at the early stage. The overall weight gain of the A4 alloy was 0.56 $mg{\cdot}cm^{-2}$. After 3 cycles, weight gain of

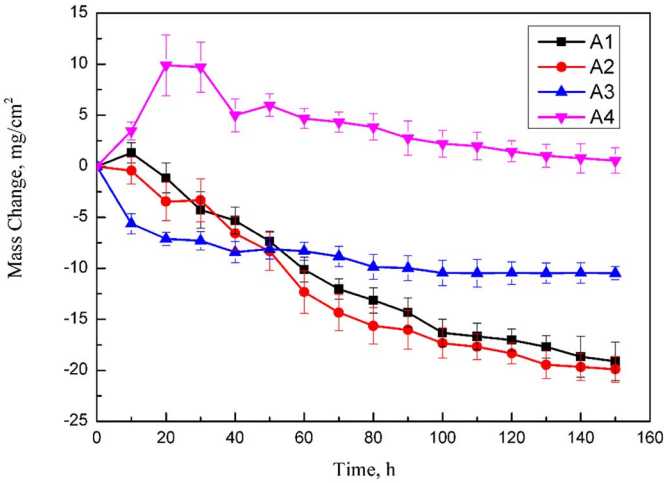


Fig. 1. NaCl-induced hot corrosion kinetic curves of A1–A4 at 750 °C.

A4 alloy was reached the peak. Then the kinetic curve of A4 alloy started to decrease. The weight change of A1 alloy was close to A2, which reached $-19\ mg{\cdot}cm^{-2}$ and $-20\ mg{\cdot}cm^{-2}$ respectively. The weight change of A3 alloy was $-10\ mg{\cdot}cm^{-2}$. And the A3 alloy showed a better stability during the hot corrosion test especially after 3 cycles. Therefore, it could be concluded that with the increase of Mo content the mass loss of the alloys decreased.

3.2. The XRD analysis

Fig. 2(a) presents the XRD patterns of the sintered porous Ni-based alloys A1–A4 alloys which consist of similar phases. The main phase was Ni-based solid solution, whose diffraction peaks were shifted to lower angles than those of pure Ni because the lattice of Ni

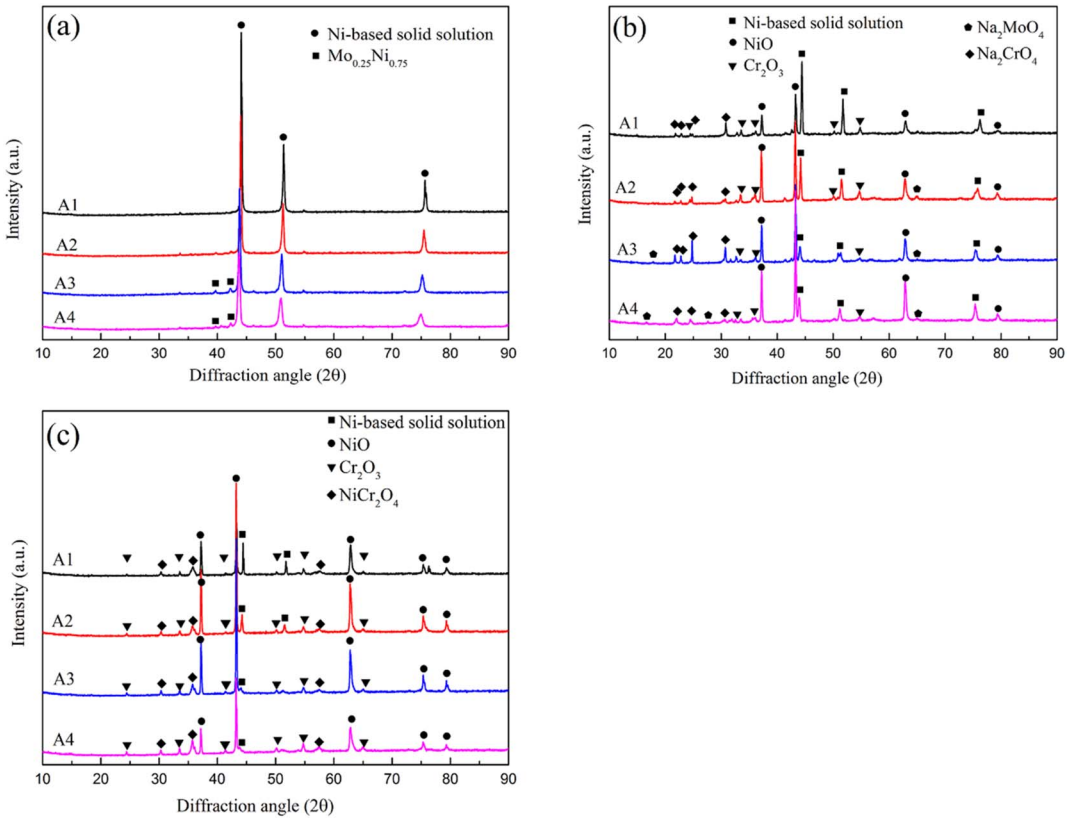


Fig. 2. XRD patterns of (a) sintering samples, (b) A1–A4 for 1 cycle hot corrosion at 750 °C, (c) A1–A4 for 15 cycles hot corrosion at 750 °C.

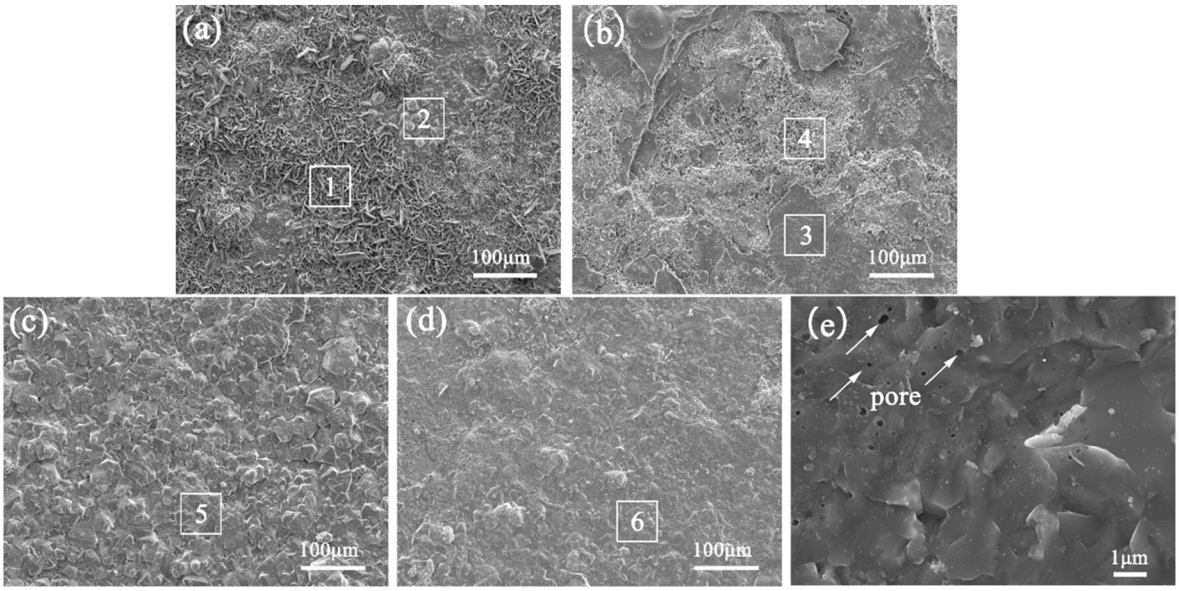


Fig. 3. Surface SEM images of (a) A1, (b) A2, (c) A3, (d) A4 alloys at low magnification for 15 cycles hot corrosion at 750 °C, (e) A4 alloys at high magnification.

was enlarged by solute strengthening elements like Cr and Mo. In addition, the peaks of $Mo_{0.25}Ni_{0.75}$ were found with the increase of Mo content (A3 and A4 alloys).

In order to elucidate the formation of the oxide scales, the surface XRD results of the four alloys after the exposure to the NaCl at 750 °C for 1 cycle are showed in Fig. 2(b). We can see that NiO, Cr_2O_3 and Na_2CrO_4 were detected in the scales. Moreover, the weak peaks of Na_2MoO_4 were also visible in A3 and A4 alloys. The XRD patterns of the corrosion products of the alloys after the hot corrosion for 150 h are showed in Fig. 2(c). All of them consist of NiO, Cr_2O_3 and $NiCr_2O_4$. From Fig. 2(c), one also can see that with the increase of Mo content, the peaks of Ni-based solid solution decreased and the peaks of NiO enhanced.

3.3. SEM/EDS analysis of exposed alloys

3.3.1. Surface SEM analysis

The SEM images showing the surface morphology of the alloys with the deposited NaCl at 750 °C for 150 h are presented in Fig. 3. The outer scales of all the alloys are shown in Figs. 3(a)–(d). Table 2 is the corresponding elemental composition measured by EDS. The A1 alloy without Mo addition showed the presence of flaky Cr_2O_3 and globular $NiCr_2O_4$ (Areas 1 and 2, as shown in Fig. 3a) after 15 cycles. The A2 alloy showed tabular oxide scales composed of Cr_2O_3 and $NiCr_2O_4$ (Area 3, as shown in Fig. 3b). Fine particles of Cr_2O_3 could be observed in Area 4, where no tabular oxide was found. The oxide scales of the A3 and A4 alloys were quite different between those of A1 and A2 alloys. Their surfaces were relatively flat covered by the dominant spinel structure of the $NiCr_2O_4$. Mo could hardly be detected except in the A4 alloy because MoO_3 could be volatilized over 700 °C [18]. At higher magnification (Fig. 3e), many small pores caused by the volatile MoO_3 or other chlorides were observed on the oxide scale of the A3 and A4 alloys after 15 cycles.

3.3.2. Cross-section SEM analysis

Cross sectional BSE (back scattered electron) images of the alloys after 15 cycles hot corrosion are shown in Figs. 4(a)–(d). It is revealed that a relatively thick corrosion layer was formed on the surface of the porous Ni-based alloys. Cr_2O_3 was detected at the edges of holes in internal of the matrices. Because the alloys had the same porous structure as coatings, O_2 and NaCl could get into the internal of alloys, therefore Cr_2O_3 was observed around the pores. It could be clearly observed that the internal oxidation of the A4

Table 2
The elemental composition on the surface of the samples after hot corrosion for 150 h (wt%).

	Ni	Cr	Mo	O
1	1.45	13.13	–	86.42
2	65.59	7.91	–	26.49
3	11.94	54.4	–	33.66
4	3.66	54.43	–	41.91
5	15.08	47.02	–	37.18
6	7.90	52.18	1.02	37.89

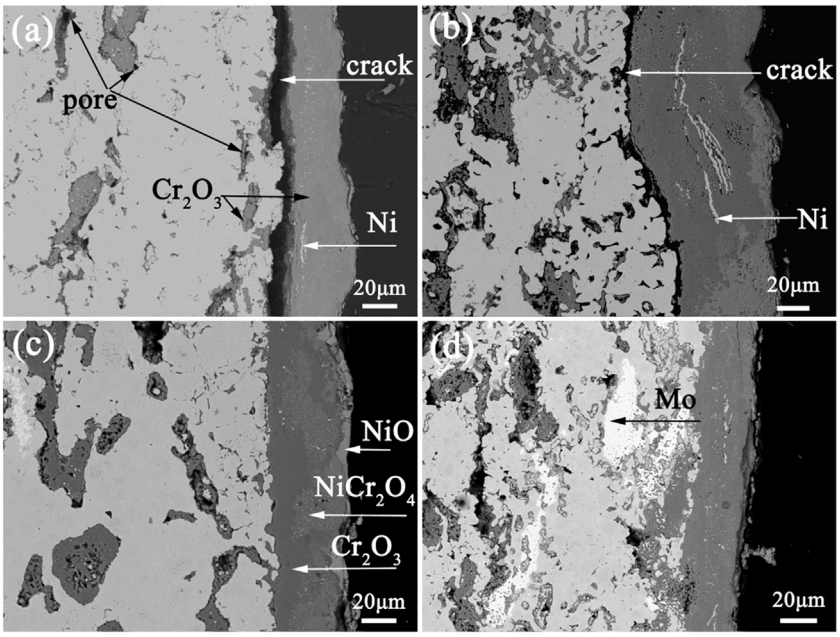


Fig. 4. Cross sectional BSE images of the (a) A1, (b) A2, (c) A3, (d) A4.

alloy was more serious compared to other alloys. This was mainly because many holes in the matrix and surface were formed due to the volatilization of MoO_3 or other chlorides, which provided channels for O_2 diffusion. This also explained why the A4 alloy in the kinetic curve of the corrosion experiments showed an obvious weight gain. The oxide scales of the A1 and A2 alloys were cracked and parted off the matrices were shown in Figs. 4(a) and (b). However, the oxide scales of the A3 and A4 featured strong adhesion of oxide scale with multi-layered structure. No cracks or voids were found in the A3 and A4 alloys, thus adding Mo is beneficial to protect the alloy. Caplan [21,22] has studied oxidation of Cr-containing alloys that in which Cr_2O_3 film was wrapped by the internal stress. He presumed that new oxidation layers might be formed on the oxide-metal interface or internal oxide film, resulting in the internal pressure stress. In addition, due to the cyclic corrosion tests of the A1–A4 alloys, the rapid-changing temperature every cycle caused high thermal stress. The cracks were easily formed during the cyclic corrosion. The A3 and A4 alloys with a certain amount of Mo were quite different from A1 and A2 alloys. It was also due to the micro porous formation of volatile MoO_3 or other chlorides release stress, so that the oxidation scale and the substrate matched very well. Wang [23] found the similar results by investigated the mechanical response of thermal barrier coatings with pores. He reported that the existing pores in the top coat would release stress concentration during the loading and unloading process. Such pores provided a channel for the diffusion of O_2 and NaCl. Ni was oxidized to NiO to form compact spinel structure of the $NiCr_2O_4$, which could prevent more efficiently chlorination and oxidation.

Elemental X-ray mapping of the cross-section of the A1 alloy is shown in Fig. 5. It could be clearly observed that Cr and O had been evenly distributed over the oxide scale. As shown in Fig. 2(c), the composition of the oxide scale was mainly Cr_2O_3 and a small amount of oxide containing Ni, Cr and O. Cr diffused to the surface and formed a protective scale during the hot corrosion, thus caused low Cr in the matrix. Compared with A1 alloy, the corrosion scale of A3 alloy was quite different. It was also observed that Ni and Cr were diffused towards the surface, as shown in Fig.6. The diffusion coefficient of Ni was higher than that of Cr in their oxides when oxygen is sufficient [24]. So Ni could diffuse to the surface of the A3 alloys due to O_2 was easily diffused into the alloys. Fig. 4(c) showed that the upper part of the scale, which mainly consists of NiO. The second layer was composed of light grey phase, which consisted of $NiCr_2O_4$. And the third layer was composed of Cr_2O_3.

3.3.3. Hot corrosion mechanism

The reactions between the elements in the alloys and NaCl at 750 ˚C may be occurred as follows:

$$Cr + 2O_2(g) + 2NaCl = Na_2CrO_4 + Cl_2(g) \quad \Delta G_{750^\circ C} = -347\ kJ \tag{2}$$

$$Mo + 2O_2(g) + 2NaCl = Na_2MoO_4 + Cl_2(g) \quad \Delta G_{750^\circ C} = -453\ kJ \tag{3}$$

$$2Ni + O_2(g) + 4NaCl = 2NiCl_2 + 2Na_2O \quad \Delta G_{750^\circ C} = 413\ kJ \tag{4}$$

$$Ni + O_2(g) + 2NaCl = NiO + Na_2O + Cl_2(g) \quad \Delta G_{750^\circ C} = 212\ kJ \tag{5}$$

One can see that Na_2CrO_4 and Na_2MoO_4 were formed in the hot corrosion, which was proved in Section 3.2 as shown in Fig. 2(b). However, the reactions (4) and (5) were unlikely to be occurred due to the positive ΔG value. It indicated that Ni did not react with NaCl during the cyclic hot corrosion. Thus a small amount Ni could be observed in the scales, as shown in Fig. 3(white phase).

The possible reactions between chlorination and oxidation are listed in Table 3. The values of Gibbs free energy for all listed

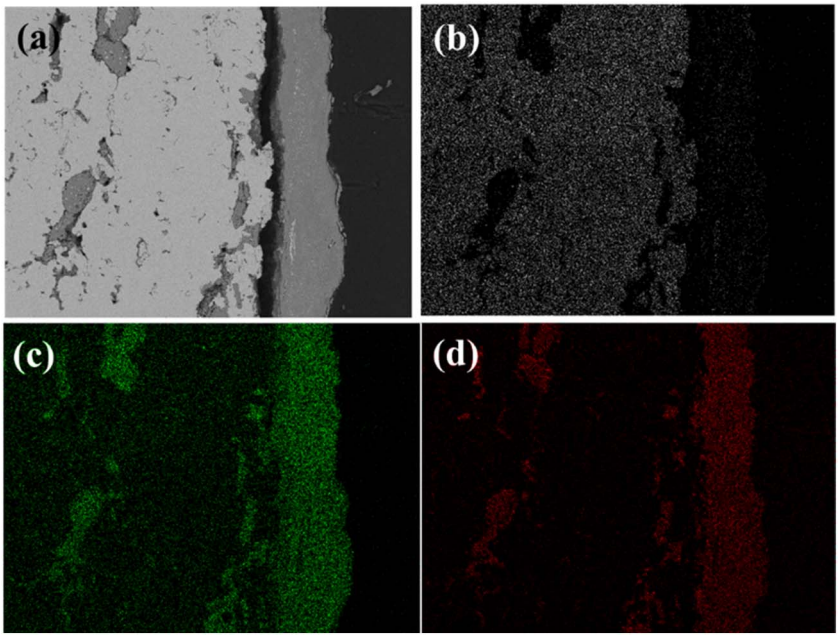


Fig. 5. BSE images and EDS maps showing the A1 after 15 cycles hot corrosion at 750 °C: (a) BSE image of A1, (b) Ni, (c) Cr, (d) O.

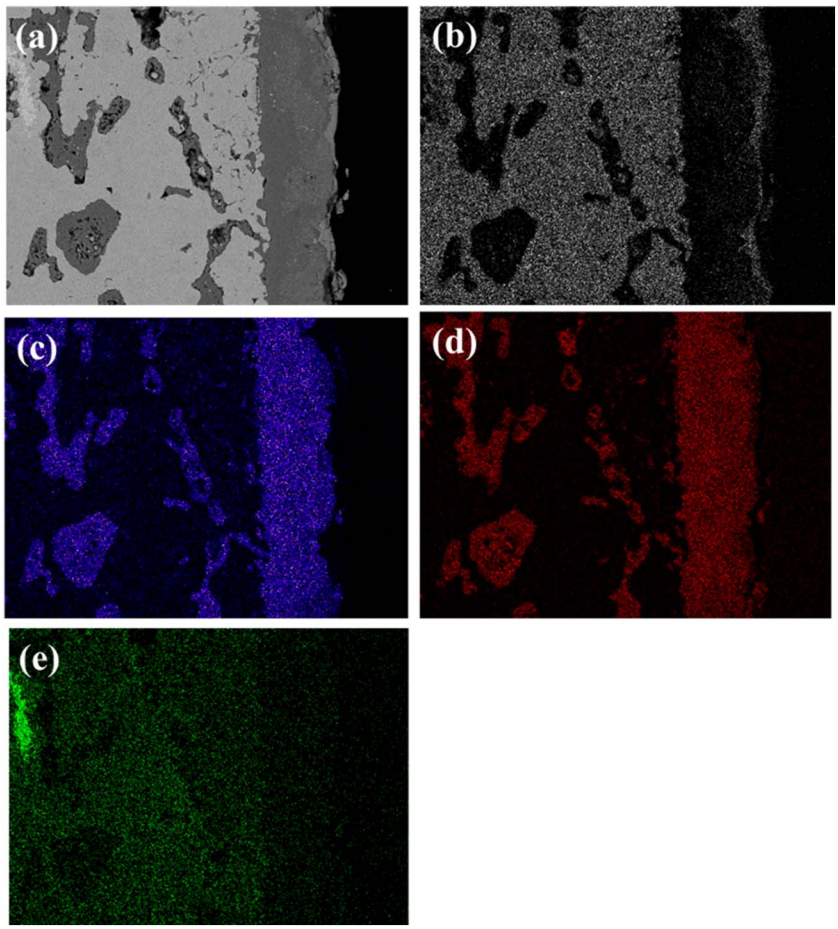


Fig. 6. BSE images and EDS maps showing the A3 after 15 cycles hot corrosion at 750 °C: (a) BSE image of A3, (b) Ni, (c) Cr, (d) O, (e) Mo.

Table 3
Possible chlorination and oxidation reactions.

Reactions	$\Delta G_{750\ ^\circ C}$/kJ
Chlorination reactions	
$Cr + Cl_2(g) = CrCl_2$	− 271
$2/3\ Cr + Cl_2(g) = 2/3\ CrCl_3$	− 227
$Mo + Cl_2(g) = MoCl_2$	− 137
$2/3\ Mo + Cl_2(g) = 2/3\ MoCl_3$	− 152
$1/2Mo + Cl_2(g) = 1/2\ MoCl_4(g)$	− 141
$Ni + Cl_2(g) = NiCl_2$	− 153
Oxidation reactions	
$4/3\ CrCl_2 + O_2(g) = 2/3\ Cr_2O_3 + 4/3\ Cl_2(g)$	− 216
$4/3\ CrCl_3 + O_2(g) = 2/3Cr_2O_3 + 2\ Cl_2(g)$	− 124
$2/3\ MoCl_2 + O_2(g) = 2/3\ MoO_3 + 2/3\ Cl_2(g)$	− 234
$2/3\ MoCl_3 + O_2(g) = 2/3\ MoO_3 + Cl_2(g)$	− 174
$2/3\ MoCl_4(g) + O_2(g) = 2/3\ MoO_3 + 4/3\ Cl_2(g)$	− 137
$2Ni + O_2(g) = 2NiO$	− 293
$4/3Cr + O_2(g) = 2/3Cr_2O_3$	− 577
$Mo + O_2(g) = MoO_2$	− 402
$Mo + O_2(g) = MoO_3$	− 488

reactions were negative, showing that all the reactions listed in Table 3 might be occurred on the basis of thermodynamics.

The schemes of NiCr and NiCrMo porous alloys during the hot corrosion induced by NaCl are illustrated in Fig. 7. It showed NaCl was deposited on the surface and then a small amount of NaCl infiltrated into the alloys via the pores. O_2 also diffused into the internal part of the alloy through the pores. Cr_2O_3 was formed around the pore (as shown in Fig. 4). The Cr was reacted to Cr_2O_3 by chlorination and oxidation. Compared with Ni, Cr could react with O to form oxides more easily. The first stage of the corrosion is shown in Fig. 7(a). A large amount of Cr was oxidized. Subsequently, Cr migrated to the surface from the substrate, resulting in decreasing of Cr content in matrix. Finally, the corrosion scale of Cr_2O_3 mixed with Ni and NiO could be formed on the surface. The corrosion process of the NiCrMo is illustrated in Fig. 7(b). Compared with NiCr, Mo in the alloy could react with NaCl to form $MoCl_2$, $MoCl_3$ and $MoCl_4$ with the melting points 1000 °C, 1300 °C and 590 °C respectively. $MoCl_4$ and MoO_3 volatilized during the corrosion test at 750 °C. The pores formed by the escape of $MoCl_4$ and MoO_3 provided a channel for the diffusion of O_2 and NaCl. Apparently, such porous structure was incapable of preventing the outward diffusion of Ni^{2+} which met the O_2 at the scale interface or surface. Then outgrown NiO was formed and reacted with Cr_2O_3, thus $NiCr_2O_4$ in the scale appeared, as shown in in Fig. 7(b). Ni was oxidized to form a NiO-$NiCr_2O_4$-Cr_2O_3 three phase corrosion scales as mentioned in Section 3.3.2. The spinel of $NiCr_2O_4$ was more compact than flaky Cr_2O_3 as shown in Fig.3. Despite the presence of Mo accelerated the oxidation of Ni, the protective scales of the $NiCr_2O_4$ formed on the surface decelerated the diffusion of NaCl and O_2 more efficiently, thus prevented further chlorination and oxidation. So

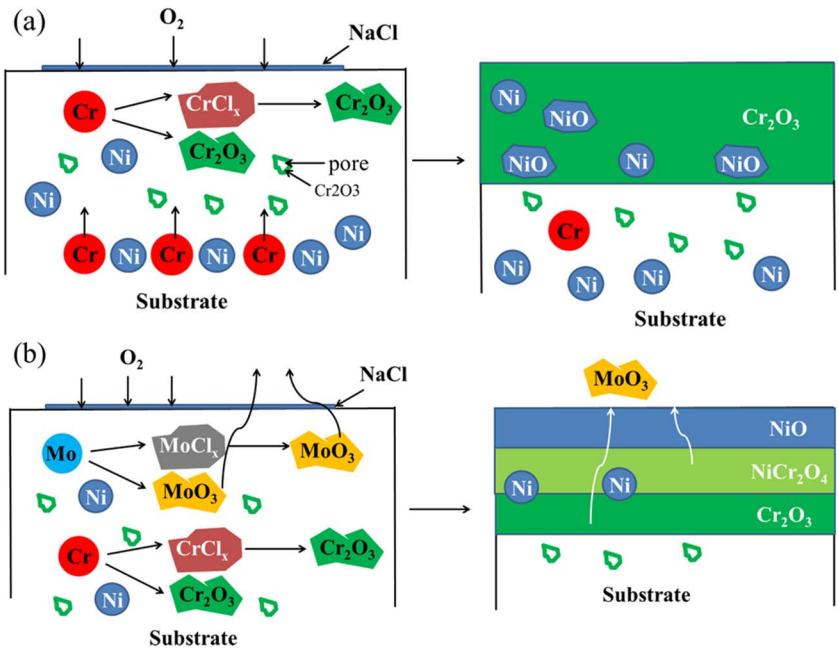


Fig. 7. Schemes of (a) NiCr porous alloy and (b) NiCrMo porous alloy during hot corrosion induced by NaCl.

the weight loss was not significant in later stages of the corrosion test for the A3 and A4 alloys, as shown in Fig. 1. However, with the increase of Mo content, the Mo oxides and chlorides in the A4 alloy were suffered from a volatilization phenomenon seriously. Such oxides and chlorides were volatile and left the scale in a gaseous form. In this way, a large numbers of pores were formed and O_2 could oxidize the elements in the substrate directly. Moreover, the corrosive Cl_2 also left the matrix via these pores. Thus oxidation reaction dominated during the corrosion process. So the oxidation weight change value of the A4 alloy was positive.

4. Conclusions

(a) Among the Ni-23Cr-xMo (0%, 4.5%, 9.0%, 13.5%, mass fraction) porous alloy, the porous Ni23Cr9Mo alloy exhibits the best stability during the hot corrosion tests.
(b) The oxide scale of NiCr without Mo addition consists of Cr_2O_3 and a small quantity of NiO and $NiCr_2O_4$. The oxide scale of the alloys containing Mo formed a NiO-$NiCr_2O_4$-Cr_2O_3 three-tier structure oxide film in the presence of NaCl.
(c) The micro porous formation owing to a volatile MoO_3 released internal stress and thermal stress. No cracks were found in the Ni23Cr9Mo and Ni23Cr13.5Mo alloys due to their better matching ability with substrates.

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